

2018-Ph-H1-Q1

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

Summary of Question:

A student calculates the vertical height a rock falls from using:

$$s = ut + \frac{1}{2}at^2$$

Given: $u = 0$, $t = 0.8 \text{ s}$, $a = 9.8 \text{ m/s}^2$

Answer:

Since $u = 0$,

$$s = \frac{1}{2} \times 9.8 \times (0.8)^2 = 0.5 \times 9.8 \times 0.64 = 3.136 \text{ m}$$



Correct answer: C. 3.1 m

Guidance for Students:

- Set $u = 0$ when object falls from rest.
- Use the correct acceleration value for gravity, 9.8 m/s^2 .

Revision Tips:

-  Memorise the SUVAT equations and know when each is applicable.
-  Watch for unit consistency (e.g. seconds, m/s^2).
-  Always round appropriately for significant figures.